

MA 210 : Real and Complex Analysis

Course Code : MA 210

Course Name : Real and Complex Analysis

L-T-P-C : 2.5-0.5-0-3

Intended for : B. Tech. in Mathematics and Computing, 2nd Year Students

Prerequisite : IC 112, IC 113

Mutual Exclusion :

Approval: 54th BoA

Course Contents

- **Analysis on $\mathbb{R}^n$:** Supremum and infimum property of the real number system, Countable and uncountable sets, Interior points and limit points, Closure of a set, Open sets and closed sets in $\mathbb{R}^n$, Bolzano-Weierstrass theorem, Compact sets in $\mathbb{R}^n$. (7 Hours)
- **Metric space:** Definition of a metric space, Examples, Open and closed sets in a metric space, Dense sets, Compact sets, Convergent sequences in a metric space, Complete metric spaces, Continuous functions between two metric spaces, Continuous functions on compact sets, Uniform continuity and uniform convergence in a metric space, Connectedness. (10 Hours)
- **Riemann integral:** Definition of Riemann integral, Examples of Riemann integrable and non-Riemann integrable functions, Some properties of Riemann integral, Continuous functions and Riemann integrability, Fundamental theorem of calculus. (4 Hours)
- **Complex Analysis:** Branch points and branch cuts, Radius of convergence of a power series, Taylor's series and Laurent's series, Classification of singularities, Poles, Picard's theorem, Zeros of analytic functions, Residue theorem, Identity/uniqueness theorem. ([7 Hours])
- **Complex (continues):** Mobius transformations, Poisson integral formula, Maximum modulus theorem, Liouville's theorem, Statement of uniqueness theorem for Dirichlet problem. (4 Hours)

Textbooks:

1. Apostol, **Mathematical Analysis**, 2nd Edition.
2. E. Kreyszig, **Advanced Engineering Mathematics**, 10th Edition.

References:

1. R. G. Bartle and D. R. Sherbert, **Introduction to Real Analysis**, 4th Edition, Wiley.
2. R. V. Churchill and J. W. Brown, **Complex Variables and Applications**, 9th Editions, 2021.
3. S. Ponnusamy, **Foundations of Complex Analysis**, 2nd Edition, Narosa, 1995.